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Propiedades de las involuciones del disco unidad

Lema 1. *Sea $w \in \mathbb{D}$ y τ_w la involución del disco unidad asociada a w, entonces*

(1) $z \in \mathbb{D} \iff |\tau_a(z)| < 1$.

(2) $z \in \mathbb{T} \iff |\tau_a(z)| = 1$.

(3) $\tau_a \in \operatorname{Aut}(\mathbb{D})$.

(4) $\tau_a \circ \tau_a = \mathrm{id}$.

***Demostración*:**

(1) Usaremos que $\forall z_1, z_2 \in \mathbb{C} : |z_1 + z_2|^2 = |z_1|^2 + 2\,\Re\left(\overline{z_1} z_2\right) + |z_2|^2$. Sea $z \in \mathbb{C}$, entonces

$$|\tau_w(z)| < 1 \iff \left|\frac{w-z}{1-\overline{w}z}\right|^2 < 1 \iff |w|^2 - \cancel{2\,\Re\left(\overline{w}z\right)} + |z|^2 < 1 - \cancel{2\,\Re\left(\overline{w}z\right)} + |w|^2\,|z|^2$$
$$\iff 1 + |w|^2\,|z|^2 - |w|^2 - |z|^2 > 0 \iff \left(1-|w|^2\right)\left(1-|z|^2\right) > 0$$
$$\overset{\substack{|w|<1 \\ \downarrow}}{\iff} 1 - |z|^2 > 0 \iff |z| < 1 \iff z \in \mathbb{D}.$$

(2) El mismo cálculo de (1) nos da que $|\tau_w(z)| = 1 \iff z \in \mathbb{T}$.

(3) Por (1), $\tau_w(\mathbb{D}) = \mathbb{D}$ y por (2), $\tau_w(\mathbb{T}) = \mathbb{T}$. Como además τ_w es una transformación de Möbius, es holomorfa e inyectiva, por lo que $\tau_w \in \operatorname{Aut}(\mathbb{D})$.

(4) $$\tau_w \circ \tau_w(z) = \tau_w\left(\frac{w-z}{1-\overline{w}z}\right) = \frac{w - \frac{w-z}{1-\overline{w}z}}{1 - \overline{w}\cdot\frac{w-z}{1-\overline{w}z}} = \frac{w(1-\overline{w}z) - (w-z)}{(1-\overline{w}z) - \overline{w}(w-z)} = \frac{z - |w|^2\,z}{1-|w|^2} = z.$$

■

Referenciado en

- `ejer-aut-disco-unidad-inversa`
- `teo-formula-aut-disco-unidad`

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